

Thanksgiving Math Crossword Solutions

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ONE DOWN

By the British Flag Theorem, $a^2 + b^2 = c^2 + d^2$ (WLOG).

Case 1: 7 & 24 exist on opposite sides of the equation

$$7^2 + b^2 = 24^2 + d^2 \text{ (WLOG)}$$

$$b^2 - d^2 = 24^2 - 7^2$$

$$(b + d)(b - d) = (24 + 7)(24 - 7) = (31)(17) = 527$$

$$\text{If } b + d = 31, b - d = 17: b = 24, d = 7 \quad \checkmark$$

$$\text{If } b + d = 17, b - d = 31: b = 24, d = -7 \quad \times$$

$$\text{If } b + d = 527, b - d = 1: b = 264, d = 263 \quad \times$$

$$\text{If } b + d = 1, b - d = 527: b = 264, d = -263 \quad \times$$

Case 2: 7 & 24 exist on the same side of the equation

$$7^2 + 24^2 = c^2 + d^2 \text{ (WLOG)}$$

$$25^2 = c^2 + d^2$$

The only integer pythagorean triples with 25 as a hypotenuse are 15-20-25 & 7-24-25.

$$(c, d) \rightarrow (15, 20) \mid (7, 24)$$

Summing the 2 Cases

$$(a, b, c, d) \rightarrow (7, 24, 7, 24)$$

$$\frac{1}{4}(a + b + c + d) = \frac{1}{4}(62)$$

$$(a, b, c, d) \rightarrow (7, 24, 15, 20)$$

$$\frac{1}{4}(a + b + c + d) = \frac{1}{4}(66)$$

Thus, the answer is $\frac{1}{4}(66 + 62) = 32$.

TWO DOWN

Let the price of the less expensive item be x , and the price of the other item be y .

$$y = 28.50$$

$$x + y = (28.50)(1 - 40\%)^{-1}$$

$$x + 28.50 = 47.50$$

$$x = 19.00$$

Thus, our answer is **19**.

THREE ACROSS

Since the holiday of Thanksgiving does not lie on a fixed date every year, it may be helpful to first see which day of the week January 1st of each year is on. For 2025, January 1st lies on a Wednesday since it exists 34 (6 mod 7) days after November 28th – a Thursday. Using the facts that non-leap years have 365 days (1 mod 7) and leap years have 366 days (2 mod 7), we can populate the table as seen on the right margin. Then, we should count the number of days between Nov. 1, 2030 and Jan. 1, 2030 to figure out what day of the week Nov. 1 lies on – January offers 30 more days after the 1st, February offers 28 days, March offers 31 days, April offers 30 days, May offers 31 days, June offers 30 days, July offers 31 days, August offers 31 days, September offers 30 days, October offers 31 days, and November offers the final day. Thus, Nov. 1, 2030 lies 304 (3 mod 7) days after Jan. 1, 2030, making November 1st a Friday. Therefore, the first Thursday of the month lies on the 7th, the second Thursday lies on the 14th, the third Thursday lies on the 21st, and the fourth Thursday lies on the 28th. Thanksgiving in the year 2030 lies on the 28th, making the answer **29**.

ONE ACROSS

$$\text{We have } a^{-1} + b^{-1} = 120^{-1}.$$

$$a + b = ab * 120^{-1}$$

$$120a + 120b = ab$$

$$ab - 120a - 120b = 0$$

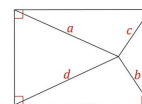
$$(a - 120)(b - 120) = 120^2$$

Since $(a - 120)$ & $(b - 120)$ can't be negative at the same time, they must both be positive.

It is easy to see that there exists a valid a for any possible value of $(a - 120)$; we are thus looking for the number of factors of 120^2 which are less than 120. Excluding 120, half the factors of 120^2 lie below 120. Since $120^2 (2^6 * 3^2 * 5^2)$ has $7 * 3 * 3 = 63$ divisors, our answer is $\frac{1}{2}(63 - 1) = 31$.

1.	3	2.	1
3.	2		9

British flag theorem



$$a^2 + b^2 = c^2 + d^2$$

Jan. 1, 2025	Wednesday
Jan. 1, 2026	Thursday
Jan. 1, 2027	Friday
Jan. 1, 2028	Saturday
Jan. 1, 2029	Monday
Jan. 1, 2030	Tuesday

AUTHOR'S NOTE

We hope you guys enjoyed these questions! Please let us know if any discrepancies in problem solutions exist by messaging in the #math-help channel of our Discord server.

<https://tinyurl.com/MEXplained>